

Areas

Physics and Mathematics of Sustainable Energy

College of the Atlantic. September 30, 2026

1. 54 kilometers is how many meters?
2. 54 square kilometers is how many square meters?
3. If a square has an area of 54 square kilometers, what is the side of the square?
4. 54 square kilometers is how many hectares?
5. 54 square kilometers is how many acres?

Windpower and Areas

Physics and Mathematics of Sustainable Energy

College of the Atlantic. October 2, 2026

1. The Hog Creek Wind Project is in Ada, Ohio.
 - (a) What is Hog Creek's nameplate capacity?
 - (b) How much electricity did it generate last year?
 - (c) What is the wind farm's capacity factor?
 - (d) What is the actual (not nameplate) power delivered by the wind farm?
 - (e) What is the area of the windfarm?
 - (f) What is the power density of the wind farm in W/m^2 ?
 - (g) The average Ohio home uses 873 kWh a month. About how many homes could the Hog Creek wind farm supply electricity to?
2. Residential electricity use in Maine is very roughly 5 million MWh/year. What area of land would be needed to generate this electricity from terrestrial windpower?
 - (a) Answer in square meters and square kilometers.
 - (b) A square of what side (in km or miles) has this same area?
 - (c) If this amount of electricity was generated using existing methods, how much CO_2 would be released into the atmosphere? Express your answer in tons per person.